

CMS Open Data H to tau tau Search: Final Analysis Note

Analysis my_analysis

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Abstract

This note documents a reduced CMS 2012 Open Data search for Higgs boson decays to tau pairs in the mu tau_h final state. The final diagnostic fit uses Run2012B/C TauPlusX reduced NanoAOD-like inputs with an integrated luminosity of 11.467/fb and a binned pyhf profile likelihood in VBF, boosted, and zero-jet categories (CMS Collaboration 2021a, 2017; Heinrich et al. 2021). The observed full-data fit gives $\mu_{\text{hat}} = 7.6642$, an observed 95% CLs upper limit $\mu < 13.3226$, and a simplified discovery diagnostic $Z = 2.9776$. The score-template validation is flagged in both the 10% data validation and full-data validation, so these numbers are open-data diagnostics rather than CMS-quality evidence for H to tau tau.

Change Log

Phase 5 v1 consolidates the Phase 4c observed result into a final analysis note and adds a PRL-style paper draft. It also adds publication-style comparison figures for the open-data diagnostic result against CMS Run 1 and Run 2 H to tau tau publications (CMS Collaboration 2014, 2018).

1 Introduction

The H to tau tau decay tests the Higgs coupling to charged leptons. CMS reported Run 1 evidence for this decay mode in JHEP 05 (2014) 104 and later observed the decay with 13 TeV data in Phys. Lett. B 779 (2018) 283 (CMS Collaboration 2014, 2018). This open-data analysis is intentionally much smaller: it uses only the reduced 2012 mu tau_h outreach samples available from the CERN Open Data Portal and does not use the full CMS calibration, embedding, or multi-channel machinery (CMS Collaboration 2021a).

The goal is a reproducible template-fit demonstration using public samples, not a substitute for the CMS publications. The result is interpreted against CMS Run 1, CMS Run 2, and ATLAS+CMS Higgs-combination context (ATLAS Collaboration and CMS Collaboration 2016), while every comparison is labelled for scope and observable differences.

2 Data And Simulation

The data are Run2012B and Run2012C TauPlusX reduced NanoAOD-like samples from the CMS Open Data H to tau tau record (CMS Collaboration 2021d, 2021e). The signal samples are ggH and VBF H to tau tau reduced samples (CMS Collaboration 2021c, 2021g). The available background simulation includes DYJetsToLL, TTbar, and W1/W2/W3JetsToLNu (CMS Collaboration 2021b, 2021f, 2021h, 2021i, 2021j).

The luminosity is 11.467/fb for the Run2012B/C TauPlusX selection following the CMS Open Data tutorial and 2012 luminosity provenance (CMS Open Data Analyses 2021; CMS Collaboration 2017). The luminosity nuisance uses 2.6% from CMS PAS LUM-13-001 (CMS Collaboration 2013). The MC event denominators come from official Open Data record `distribution.number_events` values, not from local reduced-tree entries. This avoids circular normalization by keeping data event counts out of the luminosity and MC-weight definitions.

Sample	Role	Normalization input
Run2012B/C TauPlusX	data	11.467/fb Run2012B/C luminosity
GluGluToHToTauTau	signal	sigma_prod times BR(H to tau tau) and record denominator
VBF_HToTauTau	signal	sigma_prod times BR(H to tau tau) and record denominator
DYJetsToLL	background	cross section times luminosity divided by record denominator
TTbar	background	cross section times luminosity divided by record denominator
W1/W2/W3JetsToLNu	background	cross section times luminosity divided by record denominator plus W high-mT scale

The available reduced sample list is not the full CMS analysis sample set. QCD multijet, diboson, single-top, embedded Z to tau tau, associated Higgs production, H to WW, W4/inclusive W+jets, and additional DY categories are not available as reduced samples and are documented limitations rather than silently substituted.

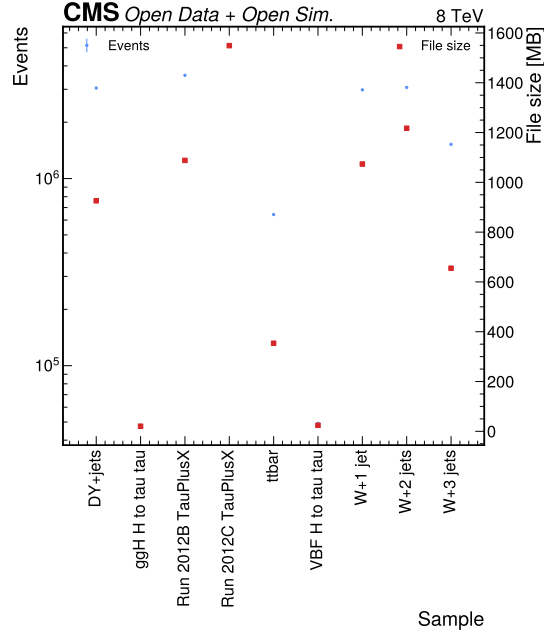


Figure 1: Sample inventory. The figure summarizes local reduced ROOT entries and file sizes for the public H to tau tau files used in the analysis. The entries are processing entries, while normalization uses official Open Data record denominators and the cited luminosity.

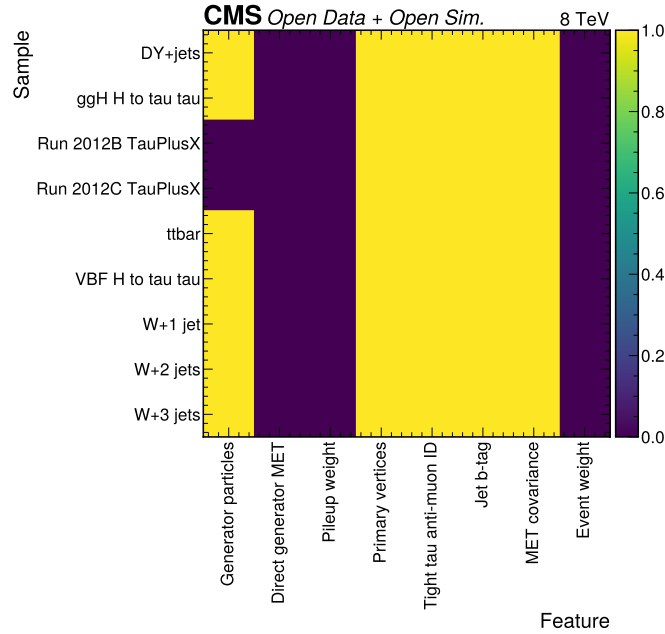


Figure 2: Branch feature availability. The figure records the reduced branch classes available in data and simulation. It shows that the core muon, tau, jet, MET, trigger, and generator-particle handles are available, while event weights, pileup weights, and direct GenMET are absent.

3 Event Selection And Categories

Events are selected with `HLT_IsoMu17_eta2p1_LooseIsoPFTau20`, matching the `TauPlusX mu tau_h` trigger path in the reduced samples. Muons require $pt > 20$ GeV, $|\eta| < 2.1$, tight identification, and finite relative isolation below 0.20. Tau_h candidates require $pt > 20$ GeV, $|\eta| < 2.3$, a decay-mode ID, tight isolation ID, tight anti-muon ID, finite isolation, and separation $\Delta R(\mu, \tau) > 0.5$.

The signal-region candidate set requires opposite-sign $\mu\tau_{\text{h}}$ pairs and muon-MET transverse mass below 40 GeV. The high- m_T W control region uses opposite-sign pairs with transverse mass above 80 GeV. A same-sign low- m_T region is retained as the available fake/QCD handle, and a Z-rich validation window is retained for DY modelling checks.

The mutually exclusive fit categories are assigned in this order: VBF first, boosted/1-jet second, and zero-jet last. VBF requires at least two clean jets, leading-dijet mass above 300 GeV, and leading-dijet $|\Delta\eta|$ above 2.5. Boosted events have at least one clean jet after failing the VBF selection. Zero-jet events have no selected clean jets.

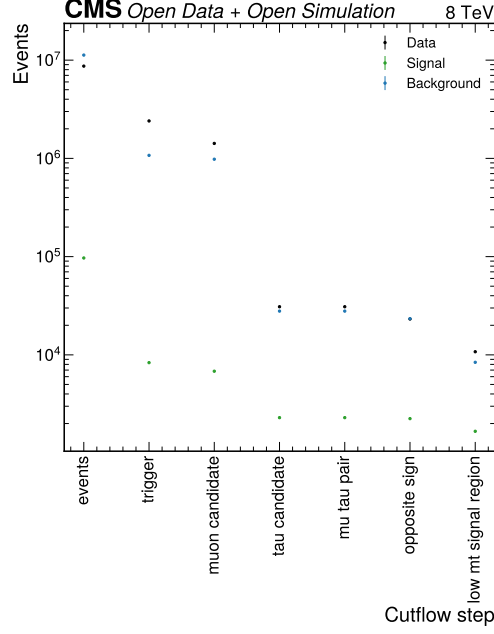


Figure 3: Cutflow summary. This plot shows raw event counts after each selection step for data, signal MC, and background MC. It verifies that the selection chain is monotonic and that the reduced samples remain populated after the trigger, object, charge, and transverse-mass requirements.

Category	Data	Background	Signal	Data/background
vbf	79	146.67	1.590	0.539
boosted	2213	2062.80	9.175	1.073
zero_jet	8466	6894.80	14.341	1.228

The category table is the final full-data selected-event summary after the W control scale is applied to W+jets. The zero-jet category dominates the event count and the combined data/background ratio, while the VBF category has the largest signal-to-background leverage but also the strongest validation deficit.

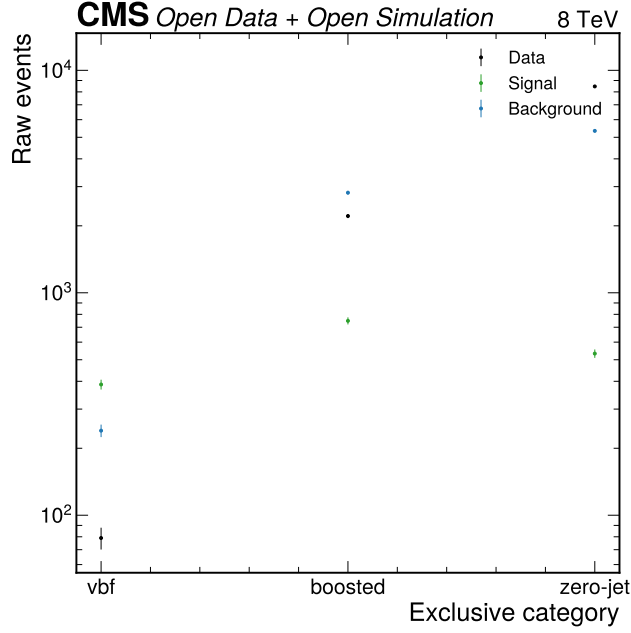


Figure 4: Exclusive category yields. The figure summarizes raw data, signal MC, and background MC counts in the VBF, boosted, and zero-jet categories. These three categories are the channels of the simultaneous fit and no event is counted in more than one fit channel.

4 Candidate Methods

The strategy evaluated visible mass, add-MET mass, an NN/MVA discriminator, and an NN missing-momentum regression idea. Visible mass and add-MET mass were implemented as transparent alternatives, but the best expected-only sensitivity came from a histogram-gradient-boosting classifier score. The NN GenMET regression was downscoped because direct GenMET and neutrino generator truth targets were not present in the reduced files.

The classifier-score method was carried forward only because it improved the expected diagnostic sensitivity in MC-only studies. That promotion was always conditional on data score-template validation. The 10% and full-data validation failed the combined ratio criterion, so the final score-template result is deliberately labelled as a diagnostic.

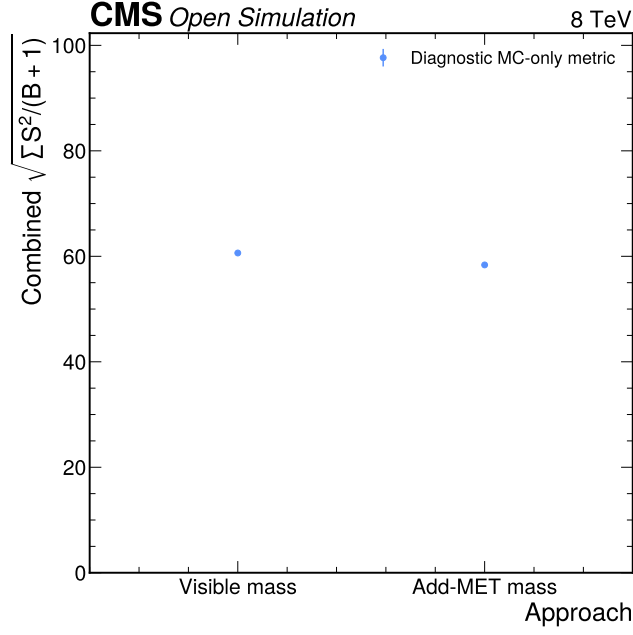


Figure 5: Approach comparison. The figure compares visible-mass and add-MET-mass approaches using a common MC-only binned separation diagnostic. It supports the existence of a transparent baseline and an add-MET cross-check, but it is not the final likelihood result.

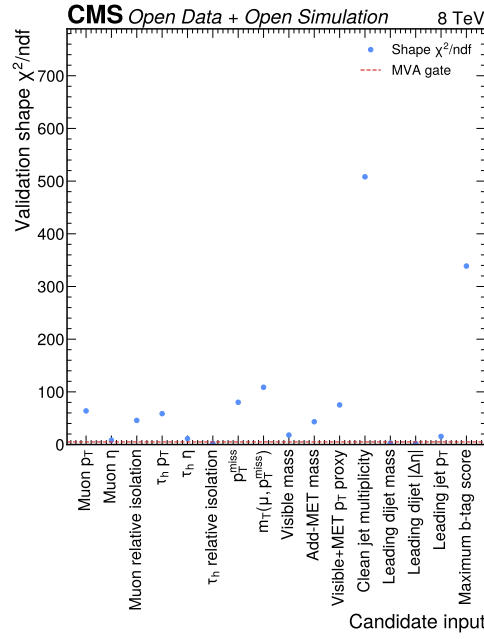


Figure 6: MVA input modelling gate. The figure shows data-versus-background-MC validation χ^2/ndf values for candidate classifier inputs. Most inputs fail the predeclared modelling threshold, which is why the classifier result is kept as an open-data diagnostic rather than a publication-quality model.

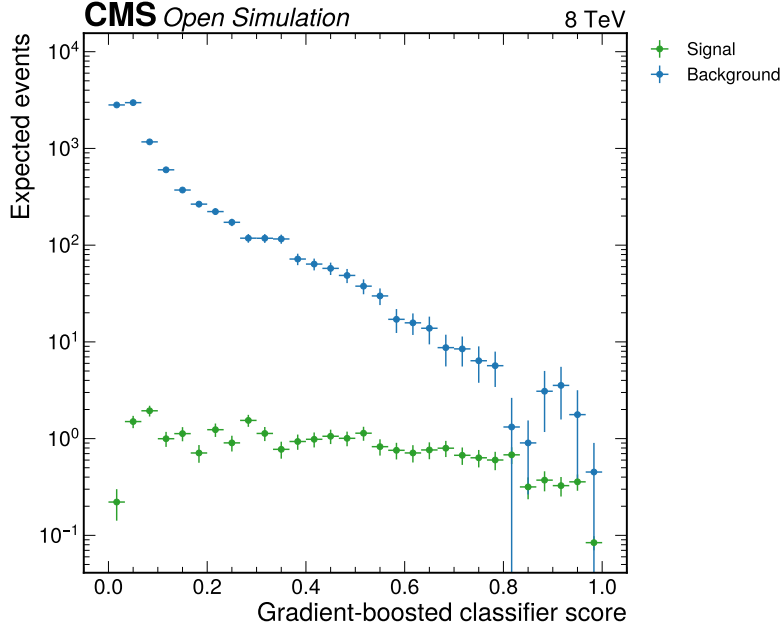


Figure 7: Expected classifier score templates. The figure shows expected signal and background score distributions after the official Open Data normalization. These templates define the fitted observable, but their full-data validation status is flagged and limits the interpretation of the final result.

5 Statistical Model

The final model is a binned profile likelihood implemented with pyhf (Heinrich et al. 2021). It has one parameter of interest, the signal-strength modifier μ , multiplying the SM H to tau tau signal templates in all fit channels. The likelihood is

$$L(\mu, \theta) = \prod_{c,b} \text{Pois}(n_{cb} \mid \mu s_{cb}(\theta) + b_{cb}(\theta)) \prod_k \pi_k(\theta_k), \quad (1)$$

where c indexes category, b indexes score bin, and θ denotes nuisance parameters. The normalization weight for a simulated background sample is

$$w_i = \frac{\sigma_i L_{\text{int}}}{N_{\text{gen},i}}, \quad (2)$$

while signal weights use the production cross section times $\text{BR}(\text{H to tau tau})$, following the Phase 3 normalization record and the Higgs cross-section context (LHC Higgs Cross Section Working Group 2017). The W+jets control scale is computed as

$$\alpha_W = \frac{N_{\text{data}}^{\text{CR}} - N_{\text{nonW}}^{\text{CR}}}{N_W^{\text{CR}}}. \quad (3)$$

The expected and observed upper limits use the modified frequentist CLs construction (Read 2002). Discovery diagnostics use the one-sided q_0 asymptotic approximation (Cowan et al. 2011). These formulae are appropriate for a compact diagnostic study, but the flagged data/MC score validation prevents interpreting the observed q_0 value as CMS-quality evidence.

6 Systematic Uncertainties

The retained nuisance model is intentionally compact because the reduced public samples lack many CMS calibration ingredients. The implemented normalization nuisances are a 2.6% luminosity uncertainty, 15% DY open-data normalization, 15% tau open-data acceptance, a W high-mT control-region normalization uncertainty, and bin-by-bin MC statistical terms. The omitted publication-level components are limitations, not hidden corrections.

Nuisance	Size	Source or derivation	Role
Luminosity	2.6%	CMS PAS LUM-13-001	MC-normalized components
DY normalization	15%	Open-data missing scale-factor limitation	DY rate
Tau acceptance	15%	Open-data missing tau trigger/ID scale-factor limitation	selected MC
W high-mT scale	4.344%	high-mT control-region count	W+jets rate
MC statistics	per bin	finite selected MC yields	all templates

The W scale improves in precision from the 10% validation to the full sample: 0.8433 ± 0.0868 at 10% and 0.8528 ± 0.0370 for the full data. The central values agree within the larger 10% uncertainty, which supports the control-region arithmetic. It does not repair the score-template validation failure.

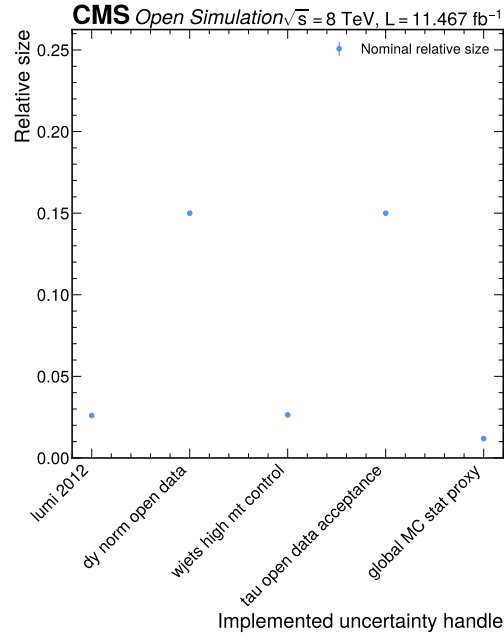


Figure 8: Expected nuisance summary. The figure shows the expected nuisance summary from the Phase 4 expected model. Its compact nuisance list reflects the reduced open-data scope and must not be read as a complete CMS systematic program.

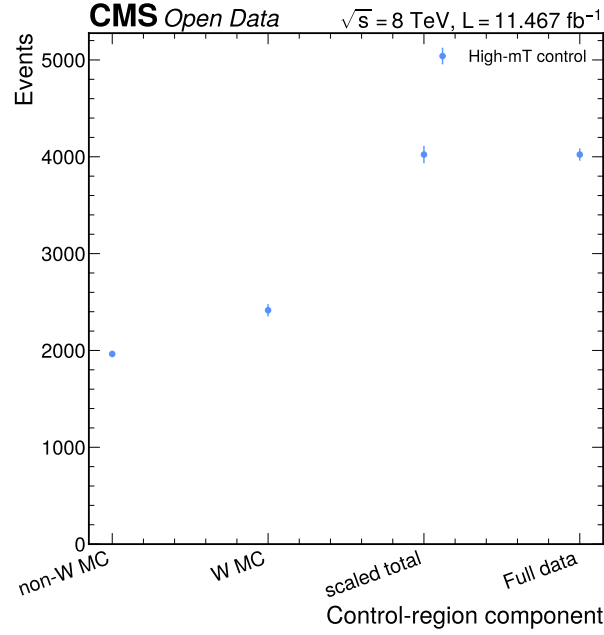


Figure 9: Full high-mT W control comparison. The figure shows the derivation inputs for the full-data W+jets control scale. The control region is outside the low-mT signal region and the scale is propagated to the observed workspace without post-unblinding signal-region tuning.

7 Expected Results

The expected-only model uses background Asimov pseudo-data, so no real signal-region data enter the expected limit or discovery diagnostic. The median expected 95% CLs upper limit is $\mu < 4.1994$, with expected band $[2.143687842430168, 2.9202283397658175, 4.199448081748937, 6.145739528239424, 8.817265131700397]$. The expected discovery diagnostic is $Z = 0.5264$.

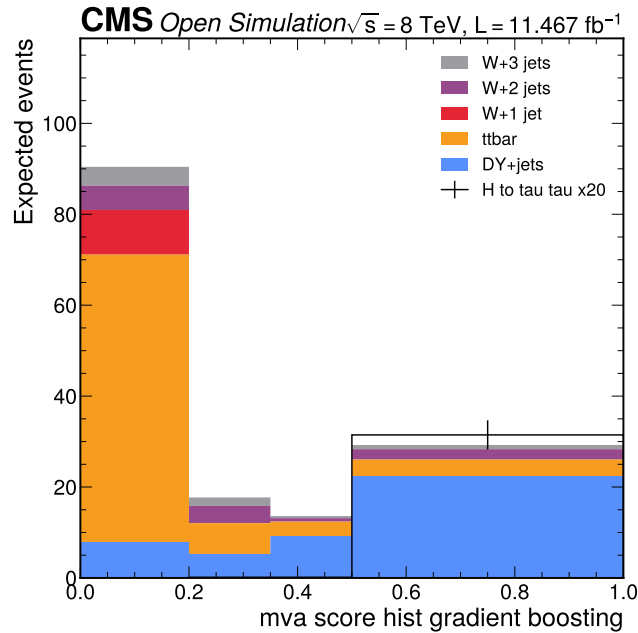


Figure 10: Expected score template in the VBF category. The figure shows expected signal and background score distributions in the VBF channel. It illustrates why the high-score region carries signal leverage, but also why sparse and mismodelled VBF data are a critical validation risk.

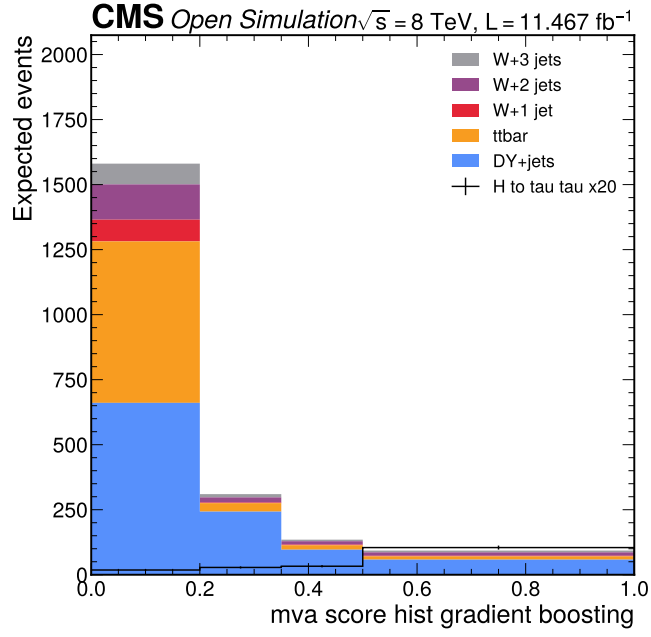


Figure 11: Expected score template in the boosted category. The figure shows the expected classifier-score template for the boosted/1-jet channel. The channel has more data than VBF and contributes a moderate validation pull in the full sample.

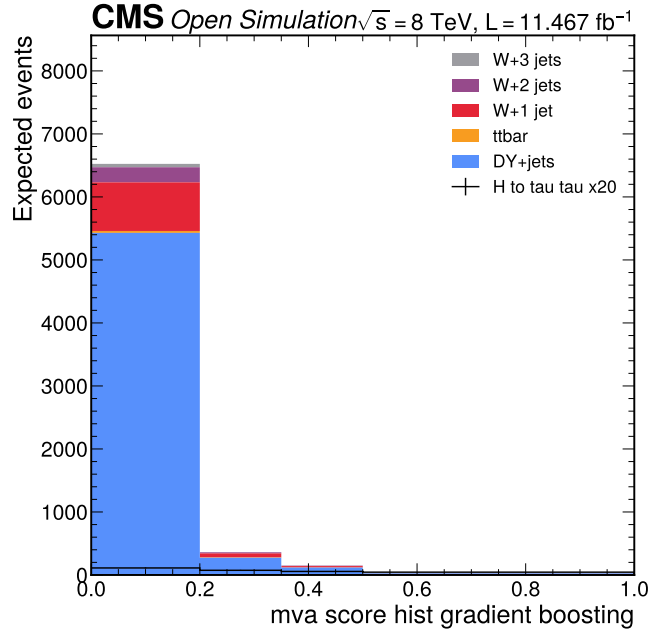


Figure 12: Expected score template in the zero-jet category. The figure shows the expected classifier-score template for the zero-jet channel. The zero-jet channel dominates event counts and drives much of the combined data/background ratio.

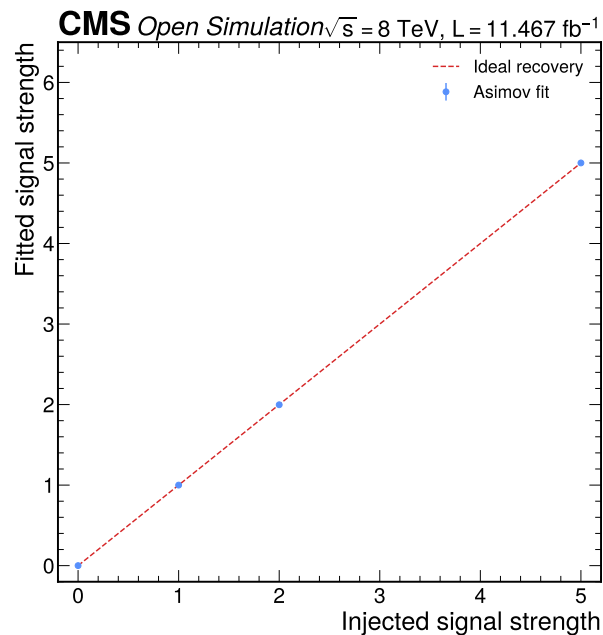


Figure 13: Signal injection recovery. The figure summarizes signal-injection recovery checks performed on pseudo-data. These checks validate model plumbing and fit response under controlled inputs, but they do not replace observed data/MC score-template validation.

8 Ten Percent Data Validation

The deterministic 10% data validation used the mask `(run * 1000003 + luminosityBlock * 9176 + event) % 10 == 0`, selecting 1120 signal-region data events. The W high- m_T scale from this subset was 0.8433 ± 0.0868 . The combined score-template data/background ratio was 1.2322 with χ^2/ndf 1.7325 and status `flagged`.

The 10% validation failed because the data/background ratio was outside the predeclared statistical criterion. This warning was carried into unblinding instead of being removed or tuned away. The full-data analysis therefore starts from a known modelling limitation.

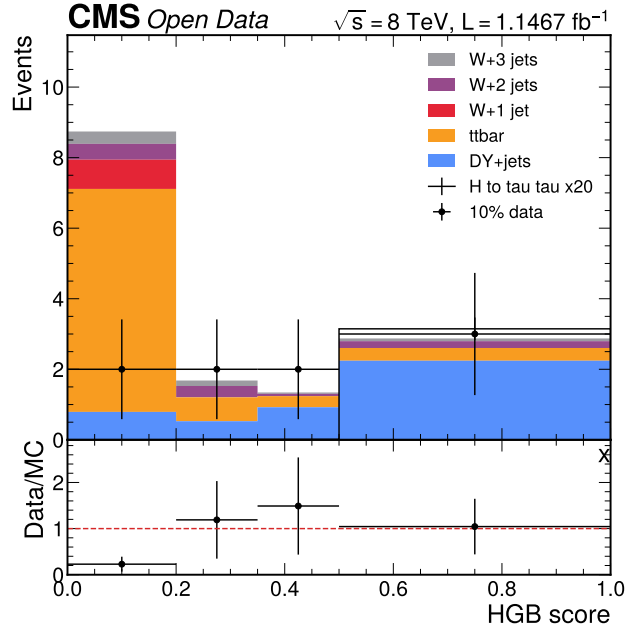


Figure 14: Ten percent score-template validation in VBF. The figure compares the 10% data subset to the frozen score-template model in the VBF category. It is one component of the validation warning that constrains the final interpretation.

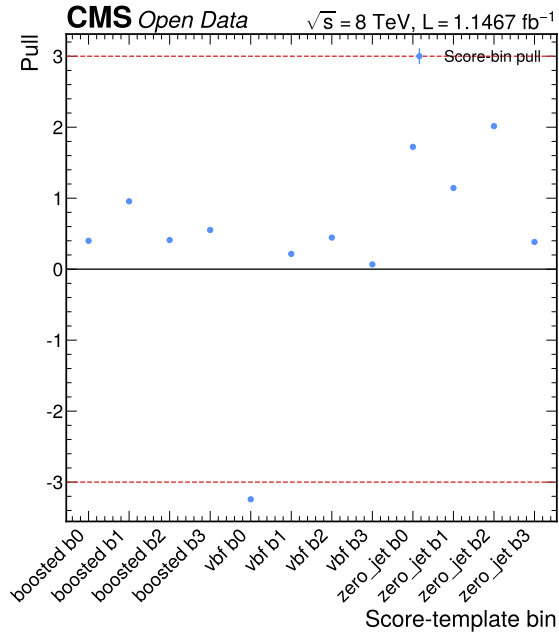


Figure 15: Ten percent pull summary. The figure summarizes 10% validation ratios and pulls. It shows that the warning is quantitative and pre-unblinding rather than an after-the-fact explanation of the full-data result.

9 Full Data Results

The full observed fit applies the frozen score-template model to all available Run2012B/C TauPlusX selected data. No selection, classifier, category, score bin, or statistical-model retuning was performed after full-data unblinding.

Quantity	Value	Interpretation
Full W high-mT scale	0.8528 ± 0.0370	Control-region scale
μ_{hat}	7.6642	Simplified open-data diagnostic
Observed 95% CLs limit	$\mu < 13.3226$	Simplified open-data diagnostic
Observed q_0 Z	2.9776	Not CMS-quality evidence
Phase 4a expected median limit	4.1994	Background-only Asimov reference
Observed/expected limit ratio	3.1725	Observed limit is much weaker than expected

The observed limit is a factor 3.17 above the median expected limit. The fitted signal strength and q_0 diagnostic are numerically large, but they occur in a model whose full-data score validation remains flagged. The correct interpretation is that the reduced open-data templates do not provide a CMS-quality evidence claim.

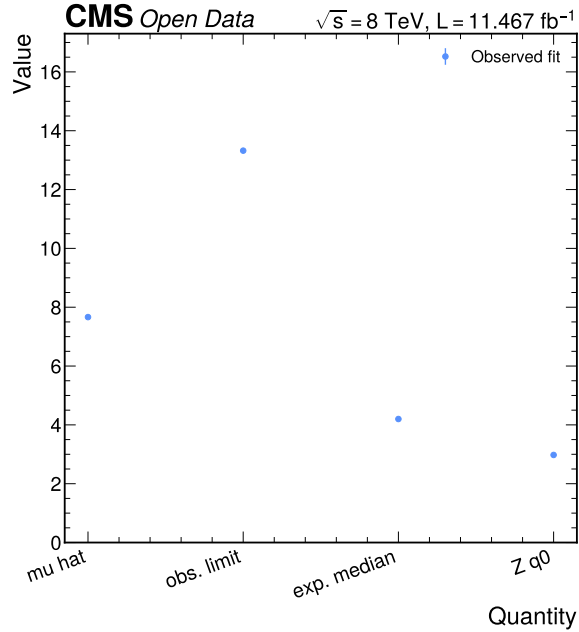


Figure 16: Observed limit and significance summary. The figure summarizes μ_{hat} , observed upper limit, expected median limit, and observed q_0 diagnostic. The caption and text deliberately label these values as simplified open-data diagnostics because the score-template validation failed.

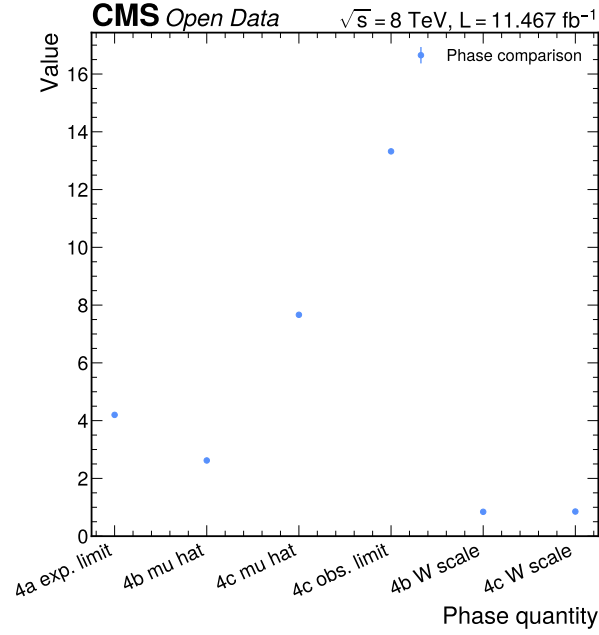


Figure 17: Comparison to expected and 10% validation. The figure compares the expected limit, 10% diagnostic fit, full-data fit, and W scale evolution. It makes the unblinding history visible and shows that the W control scale is stable while the final score-template validation remains flagged.

10 Validation Summary

Test	Data scope	chi2/ndf	p-value	Verdict	What it validates
Expected	expected-only	0.000	1.000	pass as plumbing	pyhf workspace and toy generation
Asimov toy plumbing	10% data	1.732	n/a	flagged	pre-unblinding score data/MC model
10% score templates	full data	3.218	n/a	flagged	observed score data/MC model
Full score templates	10% to full	n/a	n/a	pass arithmetic	W control-region normalization
W high-mT scale stability					

Category	Data	Background	Data/background	Chi2/ndf	Max pull	Verdict
boosted	2213	2062.80	1.073	1.687	2.14	flagged
vbf	79	146.67	0.539	4.310	3.34	flagged
zero_jet	8466	6894.80	1.228	3.656	2.47	flagged

The full combined data/background ratio is 1.1816 and the full combined chi2/ndf is 3.2176. The VBF channel has a data deficit, the zero-jet channel has a data excess, and the boosted channel is closer to the model. This category pattern is not consistent with an unqualified single signal interpretation.

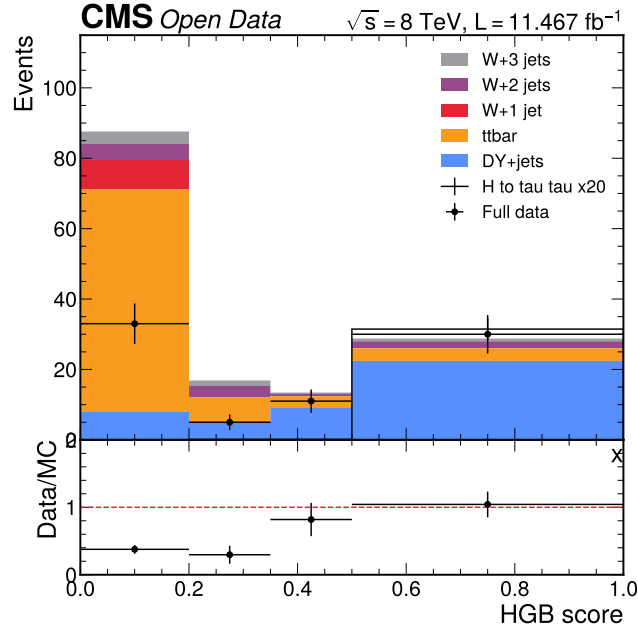


Figure 18: Full-data VBF score validation. The figure compares full data to the frozen score-template model in the VBF category. The VBF category has the largest category χ^2/ndf and a deficit relative to the background template.

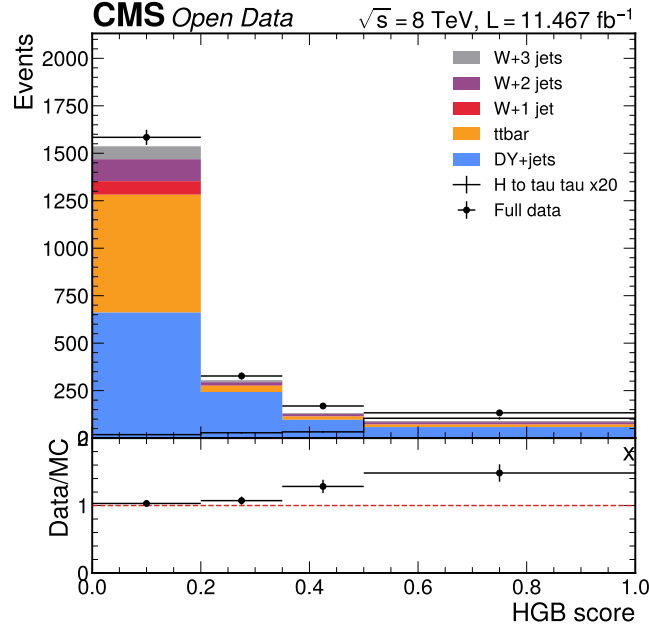


Figure 19: Full-data boosted score validation. The figure compares full data to the frozen score-template model in the boosted category. This channel has the most moderate full-data validation behavior among the three fitted categories.

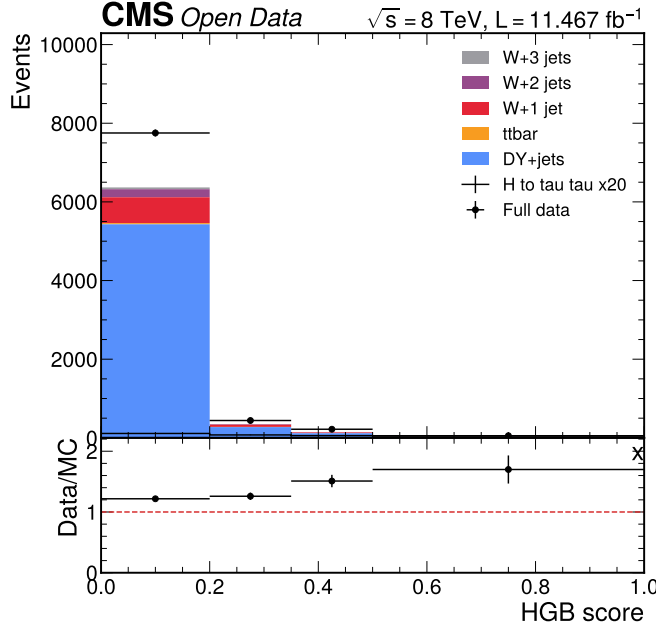


Figure 20: Full-data zero-jet score validation. The figure compares full data to the frozen score-template model in the zero-jet category. The zero-jet data excess dominates the combined data/background ratio.

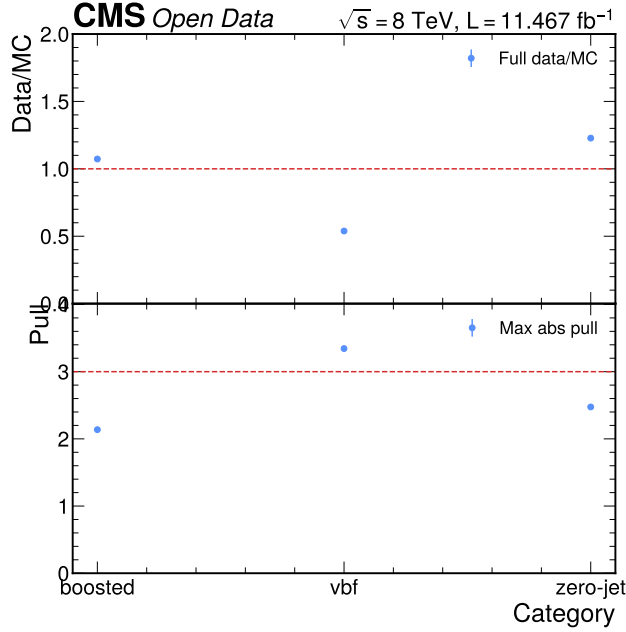


Figure 21: Observed pull and ratio summary. The figure summarizes category-level ratios, pulls, and the comparison with the 10% validation. It is the central diagnostic showing why the observed signal-strength and Z values cannot be promoted to CMS-quality evidence.

11 Published Comparisons

CMS Run 1 reported a best-fit H to tau tau signal strength of 0.78 ± 0.27 and local significance above three standard deviations for Higgs-mass hypotheses between 115 and 130 GeV (CMS Collaboration 2014). CMS Run 2 reported observed and expected significances of 4.9 and 4.7 and signal strength 1.09 with roughly 0.27 total uncertainty (CMS Collaboration 2018). These are full CMS publication results with many channels, categories, calibrations, and data-driven background methods. The open-data diagnostic fit is not directly equivalent.

Result	Scope	Significance	Signal strength or limit	Comparability
This open-data expected	8 TeV mu tau_h reduced samples	0.526	expected mu < 4.199	internal expected reference
This open-data observed	8 TeV mu tau_h reduced samples	2.978 flagged	mu_hat = 7.664; mu < 13.323	diagnostic only
CMS Run 1	7+8 TeV multi-channel	>3	mu = 0.78 ± 0.27	publication context
CMS 2018	13 TeV multi-channel	obs 4.9, exp 4.7	mu = 1.09 ± about 0.27	publication context

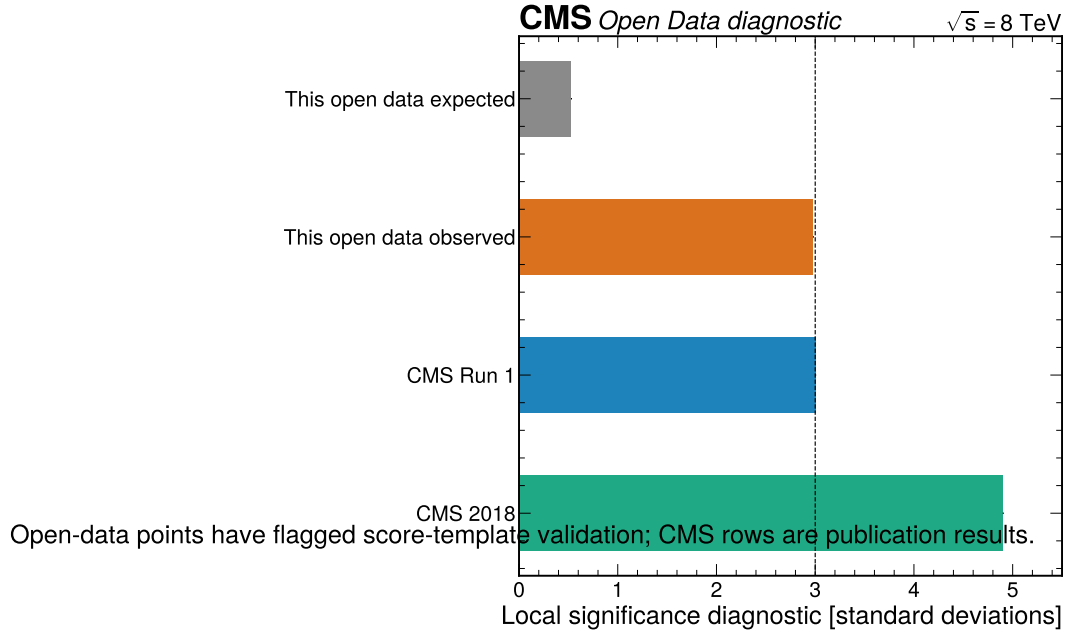


Figure 22: Significance comparison. The figure compares the open-data expected and observed diagnostics with CMS Run 1 and CMS 2018 H to tau tau publication significances. The open-data observed value is labelled as flagged and should not be read as evidence comparable to the CMS publication rows.

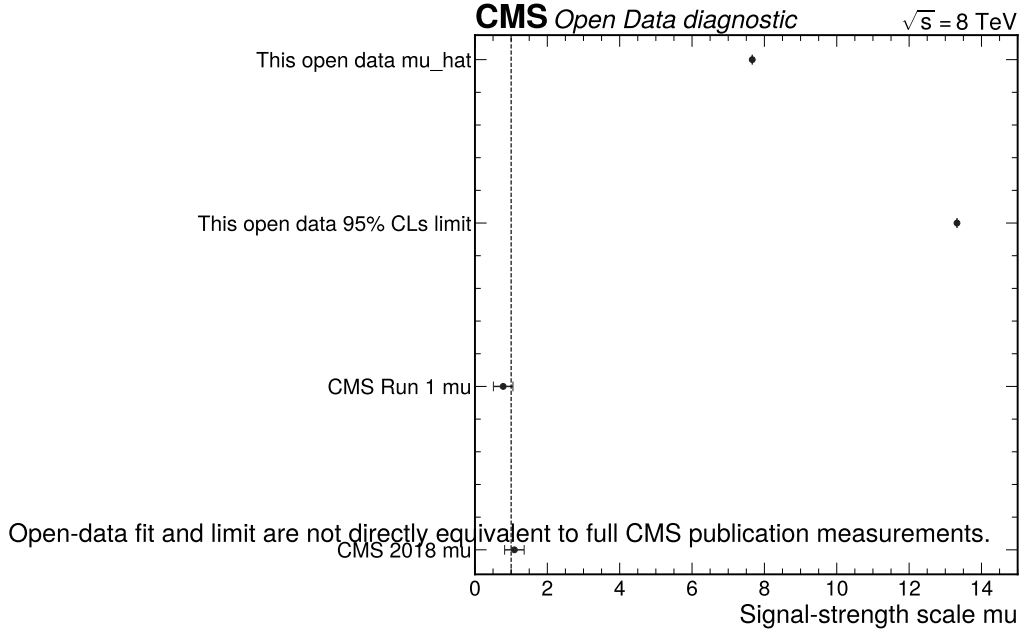


Figure 23: Signal-strength and limit comparison. The figure places the open-data μ_{hat} and observed CLs limit on the same signal-strength scale as published CMS Run 1 and Run 2 signal-strength measurements. The open-data points are not directly equivalent to the CMS measurements because of reduced channel coverage, missing calibrations, and flagged score-template validation.

12 Limitations And Resolving Power

The analysis can resolve only very large deviations from the SM expectation. The median expected 95% CLs limit of $\mu < 4.20$ already shows that an SM-strength signal is below the expected sensitivity of this reduced single-channel setup. The observed limit of $\mu < 13.32$ is still weaker because the observed score-template model is flagged.

The largest limitations are the reduced sample inventory, absent official object and trigger scale-factor machinery in the reduced files, missing QCD and electroweak background components, no pileup weights, no direct GenMET truth target, and the full-data score-template validation failure. These limitations explain why the diagnostic result sits far from published CMS-quality signal-strength measurements without requiring any claim of a new or independent physics effect.

13 Reproducibility

The final documents are generated by `phase5_documentation/src/build_phase5_docs.py`. The script reads Phase 4 JSON outputs, copies existing figures, creates Phase 5 comparison figures, merges the bibliography, writes mark-down, and compiles the TeX/PDF products. The statistical inputs are the Phase 4a expected JSON, Phase 4b partial JSON, Phase 4c observed JSON, and full observed yields JSON.

14 Conclusion

This Phase 5 pass produces a final analysis note and PRL-style draft for a reproducible CMS 2012 Open Data H to tau tau diagnostic search. The full-data fit reports $\mu_{\text{hat}} = 7.6642$, $\mu < 13.3226$ at 95% CLs, and $Z = 2.9776$. Because the 10% and full-data score-template validations are flagged, the result is explicitly not CMS-quality evidence. Its value is as a transparent, citable open-data workflow and as a documented comparison to published CMS H to tau tau analyses.

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